

Day 7 Plan

Maths: Average + Ages · Reasoning: Direction Sense · Science: Physics 7:
Electricity, Magnetism & Inventions

Today's schedule

#	Block	Time	What you do	Where
0	Day 6 recall	15 min	10 questions from Day 6, no notes. Redo yesterday's mistakes.	Page 1-2
1	Maths	75 min	Read the formulas & methods (10 min) → 2-3 book questions per pattern, timed → log mistakes	Section A
-	Break	10 min	Away from the screen	-
2	Reasoning	50 min	Learn the method (10 min) → 15 timed questions from your book → log	Section B
-	Break	10 min	Away from the screen	-
3	Science	60 min	Lessons with quick checks → 20-question practice set → answers	Section C
4	Current affairs	20 min	20 MCQs from the monthly PDF	Your PDF
5	Recall	10 min	Redo today's wrong questions; write 3 rules for tomorrow	Mistake log

Speed targets: Maths 60-70 sec per question · Reasoning 30-40 sec · Science 15-20 sec.

Block 0 — Day 6 recall (answers on the last page)

#	Question	Your answer
1	Speed of light in vacuum?	
2	Two plane mirrors at 90° form how many images?	
3	Mirror used as a vehicle's rear-view mirror?	
4	Sparkling of diamond is due to ...?	
5	Power of a convex lens with $f = 50$ cm?	
6	Lens used to correct myopia?	
7	Colour deviated least by a prism?	
8	Why is the sky blue?	
9	Why do stars twinkle?	
10	Maths: HCF 12, LCM 360, one number 72 — the other?	

Fewer than 8 right? Re-read only the lessons you missed — not the whole chapter.

A · MATHS — Average + Ages

Core idea

Average = total ÷ number of items. Most average questions are really about the **total**: convert the average to a total, change the total, then convert back. For ages, remember that **everyone gets older by the same number of years**, so the **difference** between two people's ages never changes.

Formulas

Concept	Formula / rule
Average	$\text{Sum} \div \text{count} \cdot \text{Sum} = \text{average} \times \text{count}$
Consecutive numbers / AP	$\text{Average} = (\text{first} + \text{last}) \div 2$
First n natural numbers	$(n + 1)/2 \cdot \text{first } n \text{ even: } n + 1 \cdot \text{first } n \text{ odd: } n$
One item replaced, average rises by x	$\text{New item} = \text{old item} + n \times x$
New member joins, average rises by x	$\text{New member} = \text{old average} + (n + 1) \times x$
A value was read wrongly	$\text{Correct average} = \text{old average} + (\text{correct} - \text{wrong})/n$
Average speed, equal distances	$2xy/(x + y)$
Ages in a ratio	Write ages as ak and bk ; add or subtract the same years from both

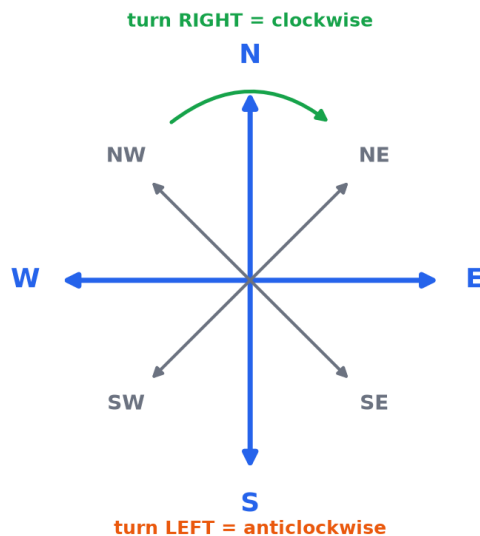
Patterns — the method and one solved example each

Pattern	Method	Solved example
1. Simple average	$\text{Sum} \div \text{count}$	10, 20, 30, 40, 50 → 30
2. Natural numbers	$(n + 1)/2$	Average of the first 50 natural numbers → 25.5
3. Replacement	$\text{Old} + n \times \text{rise}$	8 people, average rises 2.5 kg when a new person replaces one of 65 kg → $65 + 20 = \mathbf{85 \text{ kg}}$
4. Wrong entry	Adjust the total	Average of 10 numbers is 40; 64 was read as 46 → $40 + 18/10 = \mathbf{41.8}$
5. Someone joins	$\text{Old avg} + (n + 1) \times \text{rise}$	30 students, average age 15; with the teacher it's 16 → $15 + 31 \times 1 = \mathbf{46}$
6. Batting average	$\text{Total runs} \div \text{innings}$	Average 40 after 10 innings, scores 95 in the 11th → $495 \div 11 = \mathbf{45}$
7. Ages: ratio now and later	ak, bk ; add years	$A : B = 3 : 4$, after 5 years $4 : 5 \rightarrow 5(3k + 5) = 4(4k + 5) \rightarrow k = 5 \rightarrow \mathbf{15 \text{ and } 20}$
8. Ages: 'times as old'	Two equations	Father = 3 × son; in 10 years, father = 2 × son → $3S + 10 = 2S + 20 \rightarrow \mathbf{\text{son } 10, \text{ father } 30}$
9. Ages: sum + past ratio	Use k for the past	$A + B = 50$; 5 years ago $3 : 2 \rightarrow 3k + 2k + 10 = 50 \rightarrow k = 8 \rightarrow \mathbf{A \ 29, B \ 21}$

Traps

- When someone **joins**, the count becomes **$n + 1$** ; when someone is **replaced**, it stays n .
- Ages: the gap between two people never changes; a **ratio** of ages does.
- Check the answer: put the ages back into **both** conditions before marking.

B · REASONING — Direction Sense



Method: draw every move on paper, starting with North at the top. Turning **right** = 90° clockwise; turning **left** = 90° anticlockwise. For the shortest distance, use Pythagoras (common triples: **3-4-5, 6-8-10, 5-12-13**).

Facing	After turning right	After turning left	After turning back
North	East	West	South
East	South	North	West
South	West	East	North
West	North	South	East

Shadows: the Sun rises in the east, so in the **morning shadows fall to the west**; in the **evening they fall to the east**. At noon there is (almost) no shadow.

Patterns, with solved examples

Pattern	Example	Solution
Final position	Walks 5 km north, turns right and walks 3 km, turns right and walks 5 km. Where is he from the start?	N 5, E 3, S 5 → 3 km east
Shortest distance	Walks 6 km east, then 8 km north. Distance and direction from the start?	$\sqrt{(36 + 64)} = \mathbf{10 \text{ km, north-east}}$
Turns	Facing north; turns right, right, then left. Facing now?	N → E → S → East
Degrees	Facing south; turns 135° clockwise. Facing now?	South = 180°; 180 + 135 = 315° → north-west
Shadow	In the morning, Ram's shadow falls exactly to his right. Which way is he facing?	Morning shadow points west → his right is west → he faces south
Relative position	A is 10 m north of B; C is 10 m east of B. Where is A relative to C?	A is up and to the left of C → north-west

Speed rule

Draw it. Doing direction questions in your head is the main source of careless mistakes in this topic.

C · SCIENCE — Physics 7: Electricity, Magnetism & Inventions

9 lessons — the last physics chapter before Friday's revision day. The calculations are short (Ohm's law, resistances, power, bills); the facts (fuse, earthing, rules, inventors) are pure recall marks.

Lesson 1 · Charge and current (about 4 min)

Everything contains tiny charged particles. **Like charges repel, unlike charges attract.**

Charge is measured in **coulombs (C)**; one electron carries 1.6×10^{-19} C, so $1 \text{ C} = 6.25 \times 10^{18}$ electrons.

Electric current = rate of flow of charge: $I = Q/t$, unit **ampere (A).** In metal wires, electrons carry the current. By convention, current is shown flowing from + to –, **opposite** to the actual electron flow.

Conductors	Insulators	Semiconductors
Metals (silver best, then copper, aluminium), graphite, salt water	Rubber, glass, plastic, dry wood, mica	Silicon, germanium — used in chips, diodes, solar cells

Worked example — current

10 C of charge flows in 5 s $\rightarrow I = 10 \div 5 = 2 \text{ A}$.

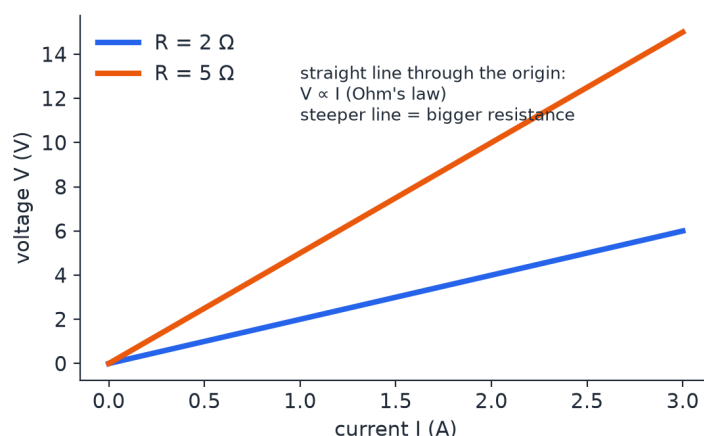
Quick check 1 — answer out loud before moving on (answers at the end)

a) A charge of 30 C flows in 1 minute. What is the current? b) Which is the best conductor of electricity?

Lesson 2 · Potential difference and Ohm's law (about 5 min)

Potential difference (voltage) is the 'push' that drives charge round a circuit: the work done per unit charge, $V = W/Q$, unit **volt (V)**. Think of water in pipes: voltage is the pressure difference, current is the flow, resistance is how narrow the pipe is.

Ohm's law: at constant temperature, the current through a conductor is proportional to the voltage across it: **$V = I R$** . R is the **resistance**, unit **ohm (Ω)**.



Worked example — Ohm's law

12 V across a $4\ \Omega$ resistor $\rightarrow I = V/R = 12 \div 4 = \mathbf{3\ A}$.

A 2 A current flows through $10\ \Omega \rightarrow V = 2 \times 10 = \mathbf{20\ V}$.

Quick check 2 — answer out loud before moving on (answers at the end)

What current flows when 220 V is applied across a $110\ \Omega$ resistor?

Lesson 3 · What resistance depends on (about 5 min)

$R = \rho L / A$ — resistance grows with **length (L)**, falls with **cross-section area (A)**, and depends on the material's **resistivity ρ** . A long thin wire resists more than a short thick one.

Material	Resistivity	Used for
Copper, aluminium	Very low	Connecting wires and cables
Nichrome (nickel-chromium alloy)	High, and doesn't oxidise easily when red-hot	Heating elements — heaters, irons, toasters
Tungsten	Fairly high; very high melting point ($\approx 3400\ ^\circ\text{C}$)	Bulb filaments

The resistance of **metals increases with temperature**.

Worked example — stretching a wire

A wire is stretched to **twice its length**. Its volume stays the same, so its area **halves**.

$R = \rho L/A \rightarrow (2L)/(A/2) = 4 \times L/A \rightarrow$ the resistance becomes **4 times**.

Exam trap

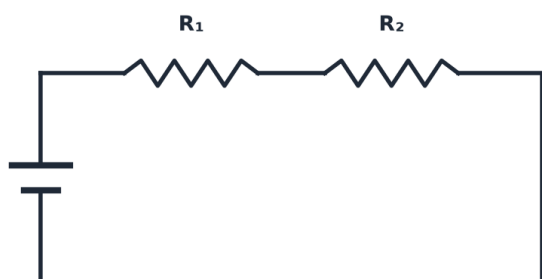
Stretching a wire to n times its length makes the resistance **n^2 times** — not n times.

Quick check 3 — answer out loud before moving on (answers at the end)

a) Which material is used for heater coils? b) A wire is stretched to 3 times its length. What happens to its resistance?

Lesson 4 · Resistors in series and parallel (about 6 min)

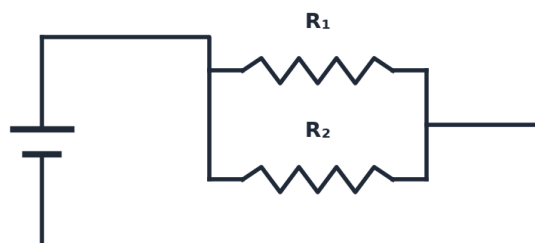
SERIES: one path



same CURRENT through each · $R = R_1 + R_2$

one bulb fails $\rightarrow$ all go off

PARALLEL: separate paths



same VOLTAGE across each · $1/R = 1/R_1 + 1/R_2$

used in house wiring: each appliance works independently

	Series	Parallel
Total resistance	$R = R_1 + R_2 + \dots$ (bigger than the biggest)	$1/R = 1/R_1 + 1/R_2 + \dots$ (smaller than the smallest)
What's the same	Current	Voltage
Two resistors	$R_1 + R_2$	$R_1 R_2 / (R_1 + R_2)$
n equal resistors R	nR	R/n

Worked example — combinations

2 Ω and 3 Ω in series = **5 Ω** ; in parallel = $6/5 = \mathbf{1.2 \Omega}$.

Three 6 Ω resistors in parallel = $6/3 = \mathbf{2 \Omega}$; in series = **18 Ω** .

Household appliances are connected in parallel, so each gets the full mains voltage and can be switched on or off independently. (Decorative serial lights are an example of series connection.)

Quick check 4 — answer out loud before moving on (answers at the end)

a) What is the resistance of 4 Ω and 4 Ω in parallel? b) Why are household appliances connected in parallel?

Lesson 5 · Electric power, energy and your bill (about 6 min)

Power $P = V I = I^2 R = V^2 / R$ (watt). **Energy** = power $\times$ time. The commercial unit is the **kilowatt-hour (kWh)** = 1 'unit' = 3.6×10^6 J.

Worked example — a 100 W, 220 V bulb

Current $I = P/V = 100 \div 220 \approx \mathbf{0.45 \text{ A}}$.

Resistance $R = V^2/P = 220^2 \div 100 = 48,400 \div 100 = \mathbf{484 \Omega}$.

Worked example — electricity bill

4 bulbs of 100 W run 5 hours a day for 30 days.

Energy = $0.4 \text{ kW} \times 5 \text{ h} \times 30 = \mathbf{60 \text{ units}}$. At ₹5 per unit $\rightarrow \mathbf{₹300}$.

A 1000 W heater running for 2 hours uses **2 units**.

Exam trap

Two bulbs (100 W and 60 W) connected in **series**: the **60 W bulb glows brighter**. It has the higher resistance, and in series the same current flows through both, so $P = I^2 R$ is larger for it. In parallel (normal house wiring), the 100 W bulb is brighter.

Quick check 5 — answer out loud before moving on (answers at the end)

A 2 kW geyser is used for 1 hour a day for 30 days. How many units does it consume?

Lesson 6 · Heating effect and electrical safety (about 5 min)

Current flowing through a resistance produces heat — **Joule's law: $H = I^2 R t$** . Used in heaters, irons, kettles, bulbs and fuses. Example: 2 A through $10\ \Omega$ for 60 s $\rightarrow H = 4 \times 10 \times 60 = \mathbf{2400\ J}$.

Safety item	Key facts
Fuse	A short wire of an alloy with a low melting point (tin-lead). Connected in series in the live wire. It melts and breaks the circuit if the current is too high
MCB	Miniature circuit breaker — a switch that trips automatically; does the same job as a fuse and can be reset
Earthing	The metal body of an appliance is connected to the earth wire, so a leaking current flows safely to the ground instead of through you
Wire colours (India)	Live: red (new: brown) · neutral: black (new: light blue) · earth: green (or green-yellow)
Short circuit vs overloading	Live touches neutral directly (very large current) · too many appliances on one socket
Mains supply in India	AC, about 220-230 V, 50 Hz

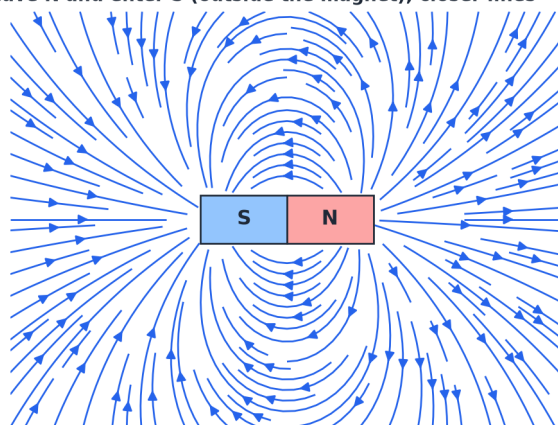
Quick check 6 — answer out loud before moving on (answers at the end)

a) What property must fuse wire have? b) In which wire is the fuse connected?

Lesson 7 · Magnets and the magnetic effect of current (about 6 min)

A magnet has a **north and a south pole; like poles repel, unlike poles attract**. A freely hanging magnet points north-south. The Earth behaves like a giant magnet — near the **geographic North Pole** lies a **magnetic south pole**, which is why a compass's north end points north.

Field lines leave N and enter S (outside the magnet); closer lines = stronger field



Oersted (1820) discovered that an **electric current produces a magnetic field**. The direction around a straight wire is given by the **right-hand thumb rule** (thumb = current, curled fingers = field). A current-carrying coil (**solenoid**) acts like a bar magnet.

Electromagnet	Permanent magnet
Coil with a soft-iron core ; magnetic only while current flows; strength can be changed	Made of steel or alloys (alnico, neodymium); keeps its magnetism
Uses: electric bells, cranes lifting scrap iron, relays, MRI machines, loudspeakers	Uses: compass needles, fridge magnets, speakers, motors

Quick check 7 — answer out loud before moving on (answers at the end)

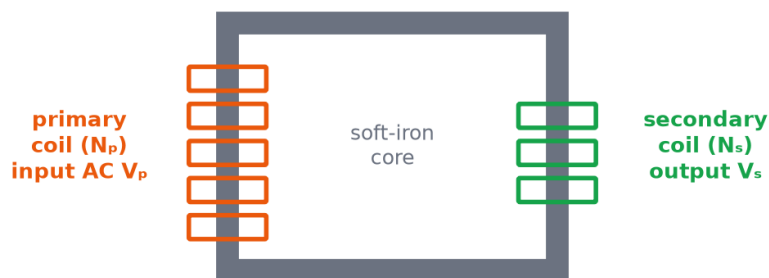
a) Who discovered that a current produces a magnetic field? b) Why is soft iron used for electromagnets?

Lesson 8 · Motor, generator, AC/DC and the transformer (about 7 min)

	Electric motor	Generator (dynamo)
Converts	Electrical → mechanical	Mechanical → electrical
Principle	A current-carrying conductor in a magnetic field feels a force	Electromagnetic induction (Faraday): a changing magnetic field induces a current
Direction rule	Fleming's left-hand rule	Fleming's right-hand rule
Examples	Fans, mixers, washing machines, pumps	Power stations, bicycle dynamo

DC (direct current) flows in one direction — batteries and cells. **AC** (alternating current) reverses direction regularly — mains supply (50 Hz in India). AC is used for supply because its voltage can be changed easily with a transformer.

more turns on the secondary → step-UP; fewer → step-DOWN



$$V_s / V_p = N_s / N_p \quad \cdot \quad \text{works only on AC}$$

Worked example — transformer

Primary 100 turns, secondary 10 turns, input 220 V → $V_s = 220 \times 10/100 = 22 \text{ V}$ (step-down, like a phone charger).

Power stations use **step-up** transformers so electricity travels at very high voltage and low current, cutting losses in the wires.

Exam trap

A transformer works **only on AC** — a steady DC current produces no changing magnetic field, so nothing is induced.

Quick check 8 — answer out loud before moving on (answers at the end)

a) Which rule gives the direction of force in a motor? b) Does a transformer work on DC?

Lesson 9 · Inventions and discoveries (high-frequency GA questions) (about 6 min)

Invention / discovery	By	Invention / discovery	By
Electric bulb (practical)	Thomas Edison	Telephone	Alexander Graham Bell
Radio	Guglielmo Marconi	Television	John Logie Baird
Electric cell / battery	Alessandro Volta	Electromagnetic induction, dynamo	Michael Faraday
Current produces magnetism	H. C. Oersted	Ohm's law	Georg Simon Ohm
X-rays	Wilhelm Röntgen	Radioactivity	Henri Becquerel
Electron	J. J. Thomson	Atomic nucleus	Ernest Rutherford
Neutron	James Chadwick	Theory of relativity	Albert Einstein
Raman effect (Nobel 1930)	C. V. Raman	Laws of motion, gravitation	Isaac Newton
Lightning conductor	Benjamin Franklin	Dynamite	Alfred Nobel
Aeroplane	Wright brothers	Analytical engine (computer)	Charles Babbage
World Wide Web	Tim Berners-Lee	Transistor	Bardeen, Brattain & Shockley
Laser (first working)	Theodore Maiman	Penicillin	Alexander Fleming

Tip: learn these in pairs out loud ('Volta — battery, Faraday — dynamo'). Two minutes a day for the next three days is enough.

Quick check 9 — answer out loud before moving on (answers at the end)

Who discovered a) X-rays, b) the neutron, c) electromagnetic induction?

Science 1-page summary — revise from this tomorrow

Topic	Must remember
Charge & current	Coulomb · $I = Q/t$ (ampere) · conventional current opposite to electron flow · silver best conductor
Ohm's law	$V = IR$ · ohm · V-I graph straight through the origin
Resistance	$R = \rho L/A$ · stretched n times $\rightarrow n^2 R$ · nichrome heaters · tungsten filaments · copper wires · rises with temperature (metals)
Combinations	Series $R_1 + R_2$ (same current) · parallel $R_1 R_2 / (R_1 + R_2)$ (same voltage) · n equal in parallel $= R/n$ · homes use parallel
Power & bills	$P = VI = I^2 R = V^2/R$ · 1 unit = 1 kWh · 100 W, 220 V bulb = 484 Ω · series: lower-watt bulb glows brighter
Heating & safety	$H = I^2 R t$ · fuse: low MP tin-lead alloy, in series in the live wire · earthing · India 220–230 V, 50 Hz AC
Magnetism	Like poles repel · field lines $N \rightarrow S$ outside · geographic N has a magnetic S pole · Oersted · right-hand thumb rule · soft-iron electromagnets
Motor / generator	Motor: elec $\rightarrow$ mech, Fleming's LEFT hand · generator: mech $\rightarrow$ elec, Faraday's EMI, Fleming's RIGHT hand
Transformer	$V_s/V_p = N_s/N_p$ · AC only · step-up for transmission, step-down for homes and chargers
Inventors	Edison bulb · Bell telephone · Marconi radio · Baird TV · Volta cell · Faraday EMI · Röntgen X-rays · Chadwick neutron · Thomson electron · Raman effect

Science practice set — 20 questions · target 6 minutes

#	Question
1	The SI unit of electric charge is the: (a) coulomb (b) ampere (c) volt (d) ohm
2	10 C of charge flows through a wire in 5 s. The current is: (a) 50 A (b) 2 A (c) 0.5 A (d) 15 A
3	Ohm's law states that: (a) $P = VI$ (b) $V = I/R$ (c) $V = IR$ (d) $R = VI$
4	A 12 V supply is connected across a 4 Ω resistor. The current is: (a) 48 A (b) 3 A (c) 0.33 A (d) 16 A
5	A wire is stretched to double its length. Its resistance becomes: (a) 2 times (b) 4 times (c) half (d) unchanged
6	The filament of an electric bulb is made of: (a) tungsten (b) nichrome (c) copper (d) iron
7	The heating element of an electric heater is made of: (a) tungsten (b) nichrome (c) copper (d) aluminium
8	The equivalent resistance of 2 Ω and 3 Ω connected in parallel is: (a) 1.2 Ω (b) 5 Ω (c) 6 Ω (d) 0.83 Ω
9	Household appliances are connected in: (a) series (b) either series or parallel (c) a single loop (d) parallel
10	The resistance of a 100 W, 220 V bulb is: (a) 220 Ω (b) 2.2 Ω (c) 484 Ω (d) 4840 Ω
11	A 1 kW heater is used for 2 hours. The energy used is: (a) 2 units (b) 1 unit (c) 0.5 unit (d) 2000 units
12	A fuse wire should have: (a) a low melting point (b) a high melting point (c) very low resistance (d) high strength
13	A fuse is connected: (a) in parallel with the live wire (b) in the earth wire (c) in the neutral wire only (d) in series with the live wire
14	The frequency of the AC mains supply in India is: (a) 60 Hz (b) 50 Hz (c) 100 Hz (d) 220 Hz
15	A transformer works on: (a) DC only (b) AC only (c) both AC and DC (d) neither
16	An electric motor converts: (a) electrical energy into mechanical energy (b) mechanical energy into electrical energy (c) chemical energy into electrical energy (d) electrical energy into heat
17	Electromagnetic induction was discovered by: (a) Michael Faraday (b) Thomas Edison (c) Alessandro Volta (d) Georg Ohm
18	The electric bulb was invented by: (a) Thomas Edison (b) Alexander Graham Bell (c) Michael Faraday (d) Nikola Tesla
19	X-rays were discovered by: (a) Henri Becquerel (b) Marie Curie (c) Wilhelm Röntgen (d) J. J. Thomson
20	The neutron was discovered by: (a) Ernest Rutherford (b) J. J. Thomson (c) Niels Bohr (d) James Chadwick

Score	Time taken	Wrong Q numbers	Mistake type (concept / formula / calc / careless / recall)
/ 20			

Answers

Science practice set

#	Ans	Why / lesson to re-read
1	a	Lesson 1
2	b	$I = Q/t$ (Lesson 1)
3	c	Lesson 2
4	b	$12 \div 4$ (Lesson 2)
5	b	Lesson 3 trap
6	a	Lesson 3
7	b	Lesson 3
8	a	$6 \div 5$ (Lesson 4)
9	d	Lesson 4
10	c	V^2/P (Lesson 5)
11	a	$1 \text{ kW} \times 2 \text{ h}$ (Lesson 5)
12	a	Lesson 6
13	d	Lesson 6
14	b	Lesson 6
15	b	Lesson 8
16	a	Lesson 8
17	a	Lesson 8–9
18	a	Lesson 9
19	c	Lesson 9
20	d	Lesson 9

Quick checks

QC	Answers
1	a) $30 \text{ C} \div 60 \text{ s} = 0.5 \text{ A}$ b) Silver
2	$220 \div 110 = 2 \text{ A}$
3	a) Nichrome b) It becomes 9 times (3^2)
4	a) $16/8 = 2 \Omega$ b) Each gets the full voltage and can be switched independently
5	$2 \text{ kW} \times 1 \text{ h} \times 30 = 60 \text{ units}$
6	a) Low melting point b) The live wire (in series)
7	a) Oersted b) It magnetises and demagnetises quickly, so the magnet switches off when the current stops
8	a) Fleming's left-hand rule b) No — only AC
9	a) Wilhelm Röntgen b) James Chadwick c) Michael Faraday

Day 6 recall (Block 0)

#	Answer	#	Answer
1	3×10^8 m/s	6	Concave
2	3	7	Red
3	Convex	8	Scattering (blue scattered most)
4	Total internal reflection	9	Atmospheric refraction
5	+2 D	10	$12 \times 360 \div 72 = 60$

Send me tonight

Day 6 recall score /10 · science practice score /20 and time · Maths and Reasoning: attempted, correct, average seconds per question · the question numbers you got wrong.